

Ethics Considerations: Interactivity and Interpretability in Data Visualization Dashboards

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Abstract

Data visualization dashboards shape how people interpret information in healthcare, business, government, journalism, etc, but the design and interaction choices embedded into these tools are not neutral. This paper examines the ethical aspects of data visualization dashboards and investigates how interactive features affect users' interpretation of time-series data compared with static charts. I will walk through the technological and societal context, then look at three areas of concern: the bias and diversity in how I evaluate users, accessibility for users with disabilities, and the bigger global and local power dynamics behind who builds, owns, and benefits from dashboard tools. I argue that visualization research carries a responsibility that most people don't talk about: when design choices change what users walk away believing, researchers aren't just studying interpretation; they are actively shaping it. I conclude by being honest about which of these concerns my project can actually address, and which it can't.

1 Introduction

My senior comprehensive project will look at how interactive features in data dashboards, things like filters, line highlighting, sorting, and drill-down, change how well users can read time-series data when compared to static charts. The mini-project found that interactive users reported much lower mental effort and found the dashboard easier to use, but their accuracy was about the same as that of static users, and they actually took longer overall [11]. The senior project will build on this by incorporating year-over-year change indicators, a possible within-subjects design, and a larger, more diverse participant pool.

On the surface, this is a fairly simple HCI project focused on a specific design question. But the bigger question with real ethical weight is: how does the way we present data change what people conclude from it? Dashboards are how doctors triage patients [13], how executives decide on staffing [1], how journalists report election results [2], and how regular people form opinions about things like climate change [14]. If interactive features change what users

believe, then researchers in this space aren't just watching from the sidelines but are part of shaping that interpretation.

This paper goes through what bias and diversity issues show up in how I'm running my evaluation, where my project falls short on accessibility, and the bigger picture of who builds dashboards and who interprets them.

2 Technological and Societal Context

2.1 Technological Context

Dashboards like Tableau, Power BI, and Looker now sit between raw data and most non-technical decision-making. These tools come with design defaults such as color choices, chart types, and interaction patterns that get reused across millions of dashboards, often without much second-guessing from the analysts using them. Past research shows that even small changes in design choices can alter which patterns users notice and which they miss [5, 7], and that interactive features aren't just for usability; they actually change how users reason about data [4, 18].

My mini project specifically used Tableau Public, the free version of Tableau, which many students, journalists, and small organizations use to build dashboards. That means the choices Tableau's design team makes, what filters look like, which interactions are easy versus hidden, which chart types they recommend, end up shaping how a whole generation of analysts learns to communicate data.

2.2 Societal Context

Dashboards became a much bigger part of everyday life during the COVID-19 pandemic, when public health agencies, news outlets, and university teams built dashboards that millions of people checked daily to decide whether to send their kids to school, visit older family members, or travel [16]. The Johns Hopkins COVID-19 dashboard alone shaped how the public understood the pandemic, and later research has shown that those readings were sensitive to framing choices: log vs. linear scales, daily vs. cumulative counts, and how the map colors were grouped [12, 9]. Same data, different design choices, different conclusions.

That's the bigger context my project sits in. The narrow research question (do filters help users interpret country-level energy data?) is part of a much bigger question about whose interpretations count, whose are made easier, and whose are made harder by the design choices built into widely used tools.

3 Bias and Diversity in User Evaluation

3.1 Sampling Bias

The biggest weakness and limitation of the mini-project was the sample: six participants, ages 17–21, mostly college students, with computer science overrepresented [11]. This is a really common problem in HCI research generally. When you recruit participants from an undergrad campus, the findings might not generalize to older, less tech-savvy, more diverse people who actually use dashboards in real life [3]. A dashboard that's "easy to use and understand" for a 20-year-old CS major who's already used Tableau is a totally different thing than one that's easy to read for a 60-year-old nurse triaging patients during a hospital shift.

For the senior comps project, I plan to recruit a larger, more demographically diverse pool of participants, but I want to be upfront that there are still limits to what an undergrad project can do here. Even 20-30 participants pulled from a single college campus are going to underrepresent older adults, people without college degrees, non-native English speakers, or folks from outside the U.S.

3.2 Whose Interpretations Count

A subtler bias question is whose interpretations get marked right when I score the tasks. My questions ask things like which country had the highest energy use in a given year, or which one increased the most, questions with mostly clear answers in the data. But Quadri et al. show that high-level visualization comprehension is actually quite interpretive: people derive meaning from data using their own background knowledge, and different interpretations aren't always wrong [10].

A participant who looks at energy data and starts thinking about it through the lens of colonial history (why do former colonial powers use so much more?) is doing something more thoughtful than someone who just spots the tallest bar, but currently, my rubric scores them the same, or even gives the second person a better score because they were faster. As a result, for the senior project, I plan to build a richer scoring method for the open-ended trend questions, perhaps one that actually credits depth of reasoning rather than just accuracy or speed alone [4, 8].

4 Accessibility

The accessibility side of this project is real, and only partly something I can address. The W3C Web Content Accessibility Guidelines outline standards for color contrast, keyboard navigation, screen reader support, and text alternatives for visuals [6]. My mini-project dashboards were built in Tableau Public, which offers some accessibility features, such as keyboard navigation, but doesn't produce fully accessible dashboards by default. Tableau's own documentation even acknowledges gaps in screen reader support for complex interactive views.

That's a real limitation. Srinivasan et al.'s work on screen-reader accessibility for dashboards argues that dashboards that prioritize the sighted lock blind and low-vision users out of data analysis; accessibility isn't an add-on but must be built in from the start [15]. The disability rights principle "Nothing About Us Without Us" [4] holds that an accessibility-respecting version of this project would involve blind and low-vision participants in designing it from the beginning, rather than being dropped into evaluation later.

Without further research on screen reader accessibility, I am unsure whether adding support for it falls within the scope of my senior comps, as I don't have the expertise or know of an appropriate participant to test it. What I can do is be upfront about my project being intended for sighted users without major impairments: choosing color palettes that meet WCAG contrast standards, ensuring interactive elements are keyboard-navigable, and acknowledging potential accessibility shortcomings.

A related issue is cognitive accessibility: users with lower data literacy or language barriers. My dashboards assume people are comfortable with line and bar charts, as well as with the underlying data concepts. That's not a given in the real world.

5 Power, Ownership, and Inequity

5.1 Who Owns The Tools

Tableau is owned by Salesforce, Power BI by Microsoft, Looker by Google. The dashboard tooling world is concentrated in a handful of big U.S. tech companies whose design decisions shape how the rest of the world interprets data. By choosing Tableau Public for my project, I'm part of that concentration: my findings about what makes dashboards interpretable are really findings about Tableau's version of interactivity, not interactivity in general. Researchers using open-source tools (D3, Vega, Plotly) might reach different conclusions because the tools themselves feel different to use.

I picked Tableau because it lets me hold the visual design constant while changing only the interactivity, which is a strength. A more thorough version of this project would replicate findings across several platforms, including open-source ones.

5.2 Global Inequity in Data and Infrastructure

The dataset I used in my mini-project comes from *Our World in Data*, which pulls data from the U.S. Energy Information Administration and the Energy Institute's Statistical Review of World Energy [17]. The data is high-quality and openly licensed, but it reflects the inequalities of where it comes from: countries with strong national statistics agencies (the U.S., U.K., E.U. members) have richer and more accurate data than countries with weaker state data infrastructure. So when you compare "energy use per capita across countries," you're implicitly comparing countries that aren't being measured equally well, and any dashboard using the data will also pass that unevenness through. This isn't a problem unique to my project, but it's not one my project solves either.

There's a second global angle too: dashboards built and refined in well-funded research labs in the U.S. shape the design vocabulary that gets exported worldwide through commercial tools. Defaults are established for English-speaking, left-to-right-reading, sighted users and become the unspoken norm for users everywhere, even for users for whom those defaults work poorly.

6 What Comps Can and Cannot Address

To be specific about scope: in the senior project, I plan to (1) recruit a bigger and more varied participant pool than the six in the mini-project, while being honest about its limits; (2) build a richer rubric for the open-ended questions that gives credit for reasoning, not just for getting the correct answer; (3) make sure my dashboards meet WCAG color contrast and keyboard navigation; and (4) report findings with clear caveats about how tied they are to Tableau specifically.

What I can't really do within this scope: meaningfully include blind and low-vision users in the design and evaluation, replicate findings across cultures and languages, fix the underlying data-quality gap that may be present in a new dataset, or get into the labor and power dynamics of how dashboards are used in workplaces.

7 Conclusion

The ethical considerations of a dashboard interpretability project are greater than they first appear. Visualizations shape how people interpret data; those understandings

shape decisions; and those decisions shape lives in areas like public health and policy, where the stakes are real. My senior comprehensive project can't solve any of these bigger concerns, but it can try to do its small piece honestly: recruit a more representative sample than the mini-project did, evaluate interpretations in ways that credit reasoning beyond accuracy, and build dashboards that meet basic accessibility standards. The most useful thing this paper can do is name the concerns the project doesn't address: global data inequity, real accessibility, and workplace power.

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